

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

English Student Edition • Focused formulas and guided practice

Formula opener

Derivative

$$\frac{d}{dx} x^n = nx^{n-1}$$

Product

$$(uv)' = u'v + uv'$$

Chain

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q001 **Written**

For $f(x) = x^2 + 2x$, find the average rate of change from -1 to 3 , and from 3 to $3 + h$.

Q002 Written

Find the average rate of change for the three source Try functions on their stated intervals.

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Fundamentals of Calculus (Limits and Derivatives)

Q003 **Written**

Find the average rate of change for all nine source Exercise functions, including fixed and h-form intervals.

Q004 Written

Find the average rate of change of a circle's area as its diameter changes from 10 cm to 18 cm.

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Fundamentals of Calculus (Limits and Derivatives)

Q005 **Written**

Find the source limits by direct substitution and by simplifying the indeterminate fraction before substitution.

Q006 Written**Find the three source Try limits, including polynomial and h-quotient forms.**

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Fundamentals of Calculus (Limits and Derivatives)

Q007 Written

Find all nine source Exercise limits, including x-limits and h-limits.

Q008 Written

If $\lim_{x \rightarrow 3} (x^2 - 5x + k)/(x - 3) = L$ is finite, find Lk .

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Q009 Written

Find the differential coefficient of $f(x) = 3x^2$ at $x = -2$ using the definition.

Q010 Written

Find the differential coefficients at the given values for the two source Try functions using the definition.

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Fundamentals of Calculus (Limits and Derivatives)

Q011 Written

Find the differential coefficients for all six source Exercise functions at their stated values.

Q012 Written

A dropped stone has displacement $s(t) = 4.9t^2$ metres. Find its instantaneous velocity at $t = 10$ seconds using the definition.

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Fundamentals of Calculus (Limits and Derivatives)

Q013 Written

Differentiate $f(x) = x^2 + 3x$ using the definition of the derivative.

Q014 Written**Differentiate the three source Try functions using the definition.**

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Fundamentals of Calculus (Limits and Derivatives)

Q015 **Written****Differentiate all nine source Exercise functions using the definition.**

Q016 Written

A regular hexagon has side length x . Write its area and differentiate it using the definition.

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Fundamentals of Calculus (Limits and Derivatives)

Q017 Written

If $f(x) = 5$ for every x except that it is undefined at $x=2$, does $\lim_{x \rightarrow 2} f(x)$ exist?

Q018 MCQ**The average rate of change from a to b is:****Choose the correct option.**

A $[f(a)+f(b)]/(b-a)$

B $[f(b)-f(a)]/(b-a)$

C $f'(a)$

D $b-a$

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Q019 MCQ

For $f(x) = x^2 + 2x$ from -1 to 3, the average rate is:

Choose the correct option.

A 2

B 3

C 4

D 8

Q020 MCQ**The average area-rate for diameter 10 to 18 is:****Choose the correct option.**A 7π B 14π C 28π D 56π

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Fundamentals of Calculus (Limits and Derivatives)

Q021 MCQ

The h-form average rate is found by:**Choose the correct option.**

- A setting $h = 1$
- B cancelling the common h
- C differentiating first
- D squaring h

Q022 MCQ**A direct-substitution limit is valid when:****Choose the correct option.**

- A the expression is defined
- B the denominator is zero
- C the numerator is infinite
- D h is negative

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Q023 MCQ

The 0/0 form requires:**Choose the correct option.**

- A factor/cancel first
- B differentiate twice
- C ignore the denominator
- D change the base

Q024 MCQ

In the finite-limit challenge, k equals:**Choose the correct option.**

A 3

B 5

C 6

D 9

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Fundamentals of Calculus (Limits and Derivatives)

Q025 MCQ

For a polynomial, the limit equals:**Choose the correct option.**

A its value at the point

B zero

C infinity

D the slope

Q026 MCQ

The definition of the differential coefficient at a is:**Choose the correct option.**

A $\lim [f(a+h)-f(a)]/h$

B $f(a+h)+f(a)$

C $[f(a)-f(0)]/a$

D $h/f(a)$

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Q027 MCQ

For $f(x) = 3x^2$, $f'(-2)$ is:**Choose the correct option.**

A -12

B -6

C 6

D 12

Q028 MCQ**The stone's velocity at 10 s is:****Choose the correct option.**

- A 49
 - B 98
 - C 196
 - D 4.9
-
-
-

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Fundamentals of Calculus (Limits and Derivatives)

Q029 MCQ

After cancelling h , take the limit as:**Choose the correct option.**A $h \rightarrow 0$ B $h \rightarrow 1$ C $x \rightarrow 0$ D $a \rightarrow \infty$

Q030 MCQ**The derivative function is:****Choose the correct option.**

- A the coefficient at one point
- B the differential coefficient for every x
- C the average rate only
- D the original function

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Fundamentals of Calculus (Limits and Derivatives)

Q031 MCQ

The derivative of $x^2 + 3x$ from the definition is:**Choose the correct option.**A $x+3$ B $2x+3$ C $2x$ D x^2

Q032 MCQ

The area of a regular hexagon of side x is:**Choose the correct option.**

- A $3\sqrt{3}x$
 - B $(3\sqrt{3}/2)x^2$
 - C $6x^2$
 - D $\sqrt{3}x$
-
-
-

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Fundamentals of Calculus (Limits and Derivatives)

Q033 MCQ

A function need not be defined at a for its limit to:**Choose the correct option.**

A exist

B be infinite only

C equal zero only

D fail

Q034 MCQ

The derivative of a constant function is:**Choose the correct option.**

A 0

B 1

C x

D the constant

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Quiz MCQ 1 [Concept Quiz](#) · [MCQ](#)

Find the average rate of change of x^2 from 1 to 3.

Choose the correct option.

A 4

B 1

C 2

D 3

Quiz MCQ 2 Concept Quiz · MCQ**Find** $\lim_{x \rightarrow 2}(x^2 + 1)$.**Choose the correct option.****A** 0**B** 5**C** 2**D** 3

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Fundamentals of Calculus (Limits and Derivatives)

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

Find $f'(x)$ for $f(x) = x^2$ using the definition.

Choose the correct option.

A 0

B 1

C $2x$

D 3

Quiz MCQ 4 [Concept Quiz · MCQ](#)

If $s(t) = 4.9t^2$, find the velocity at $t=5$.

Choose the correct option.

A 0

B 1

C 2

D 49

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Fundamentals of Calculus (Limits and Derivatives)

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the average rate of change of $3x + 1$ from 2 to 7.

Quiz WRITTEN 6 Concept Quiz · Written**Evaluate** $\lim_{h \rightarrow 0} (6 + 2h)$.

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Quiz WRITTEN 7 Concept Quiz · Written

Find the differential coefficient of $2x^2$ at $x=3$ from the definition.

Quiz WRITTEN 8 Concept Quiz · Written

State $\lim_{x \rightarrow 2} f(x)$ when $f(x)=5$ for all x except $x=2$ where it is undefined.

Answer key

Use the question number shown on each page.

Q001 4; $8 + h$	Q009 $f'(-2) = -12$	Q017 Yes; the limit is 5.	Q030 B
Q002 2; $-5/3$; $h - 6$	Q010 Substitute a into $[f(a + h) - f(a)]/h$	Q018 B	Q031 B
Q003 Apply $[f(b) - f(a)]/(b - a)$ to each printed item exactly.	cancel h , and let h tend to zero.	Q019 C	Q032 B
Q004 $7\pi \text{ cm}^2/\text{cm}$	Q011 Use the definition at each specified point and simplify exactly.	Q020 A	Q033 A
Q005 -2 ; 6	Q012 98 m s^{-1}	Q021 B	Q034 A
Q006 Substitute after removing any common factor; polynomial limits are obtained directly.	Q013 $f'(x) = 2x + 3$	Q022 A	Quiz MCQ 1 A
Q007 Each limit is obtained by direct substitution or cancellation of the factor causing $0/0$.	Q014 Expand each $f(x + h) - f(x)$, cancel h and take the limit.	Q023 A	Quiz MCQ 2 B
Q008 6	Q015 Apply $f'(x) = \lim_{h \rightarrow 0} [f(x + h) - f(x)]/h$ to every printed polynomial.	Q024 C	Quiz MCQ 3 C
	Q016 $A = \frac{3\sqrt{3}}{2}x^2$, $A'(x) = 3\sqrt{3}x$	Q025 A	Quiz MCQ 4 D
		Q026 A	Quiz WRITTEN 5 3
		Q027 A	Quiz WRITTEN 6 6
		Q028 B	Quiz WRITTEN 7 12
		Q029 A	Quiz WRITTEN 8 5